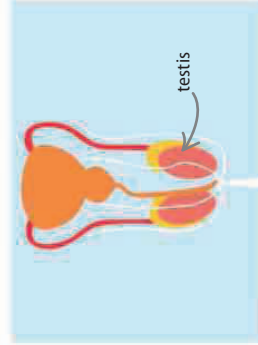
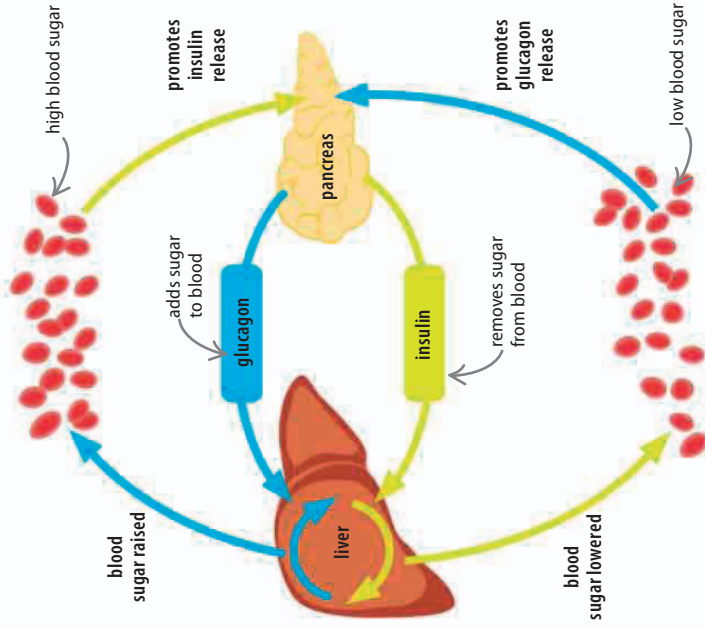


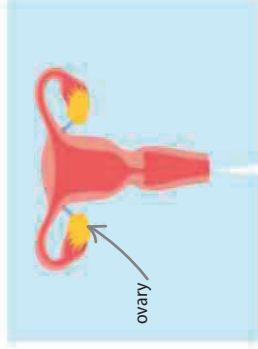
◁ Insulin

This hormone is produced by the pancreas. Its main job is to convert the sugar that has entered the blood following a meal into a starchy fuel store called glycogen. This takes place in the liver. If blood sugar drops, another hormone, called glucagon, reverses the process, turning the glycogen back into sugar to fuel the body.



△ Testosterone

The male hormone is produced by the testes (singular: testis), the male reproductive organs. As well as controlling the production of sperm, testosterone makes the body develop male characteristics, such as increased body hair and larger muscles. Testosterone also increases the willingness to fight (although it does not make you any better at it).



△ Estrogen

This is a female hormone produced by the ovaries. It is involved in the production of eggs, making them ready for reproduction on a monthly cycle. Estrogen is also responsible for making the body develop secondary sexual features, such as mammary glands and pubic hair during puberty.

Thermoregulation

The human body maintains a constant body temperature so its metabolism runs at a constant rate. As a result, the body must conserve its heat in cold conditions and shed excess heat in warm ones. Its processes for keeping its body temperature steady are called thermoregulation.

▽ Hot and cold

Thermoregulation makes use of the basic principles of heat transfer. Heat is lost via skin flooded with warm blood, and also by the evaporation of sweat secreted on the skin. When cold, the body curls up to decrease its surface area and so reduce heat loss.

Hot conditions	Cold conditions
Vasodilation The blood vessels in the skin widen to allow warm blood to radiate heat into the air.	Vasoconstriction The blood vessels contract, so less blood reaches the skin, reducing heat transfer.
Sweating As sweat evaporates, it takes some of the body's heat with it.	Shivering The rapid movement of muscles when shivering generates heat.
Pilorelaxation Body hair lies flat, allowing cool breezes to get close to the skin.	Piloerection (goosebumps) Body hair stands up, to keep a layer of warm air next to the body.
Stretching out Moving around allows heat to be lost from a larger surface area.	Curling up Curling tight reduces the surface area losing heat.

In **very cold water**, a human's heart rate slows and blood is sent only to the brain and vital organs to conserve oxygen—so the person can survive for several minutes without **breathing**.